

Introduction to Physics

AI Tutor - Physics - Lesson 1

What is Physics?

Physics is the natural science that studies matter, energy, motion, and the forces that govern how objects behave. It seeks to explain everything from why an apple falls to the ground to how galaxies move through space.

Major Branches of Physics

- **Mechanics** - the study of motion and forces.
- **Thermodynamics** - the study of heat and energy transfer.
- **Electromagnetism** - the study of electricity, magnetism, and light.
- **Modern Physics** - relativity and quantum mechanics.

The Scientific Method

Physics relies on observation, forming a hypothesis, testing it through experiments, and refining theories based on evidence. This method allows physicists to build reliable, predictive models of how nature works.